SQL 2

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ISIN 325 - Database Security

Introduction

In this assignment for ISIN 325, we were tasked with creating 12 different queries for SQL Server to gather data, delete records, and create table joins and stored procedures.

1. Display the count of the number of rows in a table.

```
Select count(*) as ProviderCount from Provider
```

118 % No issues found

Results | Messages

	ProviderCount
1	27

2. Delete a specified record in a table and prove that it is deleted.

2-IN-1\SQLEX....ProviderType | SQLQuery1.sq...\wills (51))* | SQL2 assignme...-1\wills (67))

```
Select count(*) as ProviderTypeCount from ProviderType --checking the number of records before delete
Delete from ProviderType --deletes the records
Where ProviderTypeID = 5 or ProviderTypeID = 6
Select count(*) as ProviderTypeCount from ProviderType --checks the records are deleted
```

118 % No issues found Ln: 4,

Results | Messages

	ProviderTypeCount
1	6

	ProviderTypeCount
1	4

3. Two table join that creates a view.

```sql
Create view vw_GetProviderAndType as --create view and name
Select ProviderID, FName, LName, ProviderType
From Provider P, ProviderType PT
Where P.ProviderTypeID = PT.ProviderTypeID

go
Select * from vw_GetProviderAndType --shows the view
```

118 % 1 0

Results Messages

	ProviderID	FName	LName	ProviderType
1	1	James	Wilson	Physician
2	2	Emily	Carter	Registered Nurse
3	3	Michael	Reed	Physician
4	4	Sarah	Nguyen	Physical Therapist
5	5	David	Patel	Physician
6	6	Laura	Hernandez	Registered Nurse
7	7	Robert	Kim	Phlebotomist
8	8	Jessica	Morris	Physical Therapist
9	9	Daniel	Lopez	Physician
10	10	Amanda	Brooks	Registered Nurse
11	11	Matthew	Sanders	Physician
12	12	Olivia	Price	Physical Therapist
13	13	Andrew	Bell	Phlebotomist
14	14	Stephanie	Rivera	Registered Nurse
15	15	Christopher	Howard	Physician
16	16	Nicole	Ward	Physical Therapist
17	17	Brian	Cox	Physician
18	18	Rachel	Peterson	Registered Nurse
19	19	Kevin	Flores	Phlebotomist
20	20	Megan	Russell	Physical Therapist
21	21	Jonathan	Long	Physician
22	22	Hannah	Ramirez	Registered Nurse
23	23	Eric	Powell	Phlebotomist
24	24	Victoria	Bennett	Physical Therapist

4. A three-table join qualified by a key value from one view and one other table.

2-IN-1\SQLEX....ProviderType | SQLQuery1.sq...\wills (51))* | SQL2 assignme...1\wills (67))*

```
Select LName, ProviderType, Ailment
From vw_GetProviderAndType V, PatientAilment PA, Ailment A
Where V.ProviderID = PA.ProviderID
and A.AilmentID = PA.AilmentID
```

118 % 3 0

Results | Messages

	LName	ProviderType	Ailment
1	Wilson	Physician	Hypertension
2	Reed	Physician	Migraine
3	Carter	Registered Nurse	Diabetes Mellitus
4	Patel	Physician	Seasonal Allergies
5	Nguyen	Physical Therapist	Asthma
6	Hernandez	Registered Nurse	Chronic Back Pain
7	Kim	Phlebotomist	Depression
8	Morris	Physical Therapist	Eczema
9	Lopez	Physician	Arthritis
10	Brooks	Registered Nurse	Irritable Bowel Syndrome
11	Sanders	Physician	Anxiety Disorder
12	Price	Physical Therapist	Vertigo
13	Bell	Phlebotomist	Coronary Artery Disease
14	Rivera	Registered Nurse	Chronic Obstructive Pulmonary Disease
15	Howard	Physician	Seasonal Allergies
16	Ward	Physical Therapist	Influenza
17	Cox	Physician	Migraine
18	Peterson	Registered Nurse	Sleep Apnea
19	Flores	Phlebotomist	Hypertension
20	Russell	Physical Therapist	Anemia
21	Long	Physician	Pneumonia
22	Ramirez	Registered Nurse	Urinary Tract Infection
23	Powell	Phlebotomist	Gastroesophageal Reflux Disease
24	Bennett	Physical Therapist	Osteoporosis
25	Coleman	Physician	Chronic Kidney Disease
26	Wilson	Physician	Hypothyroidism

5. Use five SQL functions (such as SUM). This should be five queries with at least one function for each query, and not the same functions as you used in previous assignments.

```sql
-- 1. Returns the ASCII value of the first letter from each patients last name
Select ASCII(LName) as PTName
From Patient

-- 2. Extracts the last three letters of each patients last name
Select RIGHT(LName, 3) as Last3Patient
From Patient

-- 3. returns the length of each patients first name
Select LEN(FName) as LengthPatient
From Patient

-- 4. Returns an integer as a string
Select STR(ProviderID) as ProviderIDString
From Provider

-- 5. Returns the number of bytes in the provider first na
Select DATALENGTH(FName) as ProviderNameBytes
From Provider
```

118 % No issues found

Results Messages

	PTName
1	83
2	74
3	66
4	84
5	65

	Last3Patient
1	ith
2	son
3	own
4	lor
5	son

	LengthPatient
1	4
2	4
3	6
4	8
5	7

	ProviderIDString
1	1
2	2
3	3
4	4
5	5
6	6

	ProviderNameBytes
1	10
2	10
3	14
4	10

6. Create a SQL stored procedure to insert values into a row using variables passed in on execution.

```sql
Create Proc spu_InsertProviderTypeDynamic --creates the stored procedure
 @ProviderTypeID int, @ProviderType nvarchar(50), @ProTypeDesc nvarchar(Max)
as

Insert into ProviderType
    (ProviderTypeID, ProviderType, ProTypeDesc)
Values
    (@ProviderTypeID, @ProviderType, @ProTypeDesc)
go
EXEC spu_InsertProviderTypeDynamic --inserts the values into the table
    5,
    'CardoVascular Surgeon',
    'Heart Doctor'

exec sp_depends 'ProviderType' --shows the stored procedures
```

118 % 2 0

Results | Messages

	name	type
1	dbo.spu_InsertProviderType	stored procedure
2	dbo.spu_InsertProviderTypeDynamic	stored procedure
3	dbo.vw_GetProviderAndType	view

7. Write the SQL to create a table, key, and index. This must not be generated SQL. Example after #12.

```sql
--Creates the table
CREATE TABLE Student

--Unique Identifier
(StudentID int NOT NULL,
FName nvarchar(20) NOT NULL,
LName nvarchar(30) NOT NULL,
MName nvarchar(20) NULL,
CONSTRAINT PK_Student PRIMARY KEY CLUSTERED
(StudentID ASC)
ON [PRIMARY]) -- key
ON [PRIMARY] -- index
```

8. Create a query that uses Group By

```sql
Select LName, Count(PatientID) as PatientCnt
From Provider P, PatientAilment PA --Uses two differenty tables to compile records
Where P.ProviderID = PA.ProviderID
Group By LName
Order by PatientCnt Desc --Sorts the records in decending order by last name
```

118 % 2 0

Results Messages

	LName	PatientCnt
1	Cox	2
2	Flores	2
3	Hernandez	2
4	Howard	2
5	Kim	2
6	Long	2
7	Lopez	2
8	Morris	2
9	Nguyen	2
10	Patel	2
11	Peterson	2
12	Price	2
13	Ramirez	2
14	Reed	2
15	Rivera	2
16	Russell	2
17	Sanders	2
18	Ward	2
19	Wilson	2
20	Bell	2
21	Brooks	2
22	Carter	2
23	Coleman	1
24	Bennett	1
25	Powell	1

9. Demonstrate the use of distinct, ascending, descending in a query. The result set must have at least 7 rows.

```sql
--Distinct
Select Distinct LName
From Provider
Order by LName
--Ascending
Select LName
From Provider
Order by LName
--Descending
Select LName
From Provider
Order by LName Desc
```

118 % No issues found

Results Messages

	LName
1	Bell
2	Bennett
3	Brooks
4	Carter
5	Coleman

	LName
1	Bell
2	Bennett
3	Brooks
4	Carter
5	Colem...
6	Cox

	LName
1	Wilson
2	Ward
3	Stev...
4	Smith
5	Sand...

10. Create a query that demonstrates an inner join (includes join in statement)

```
Select LName, ProviderType
from Provider P
    join ProviderType PT
    on P.ProviderTypeID = PT. ProviderTypeID
```

118 % No issues found

Results Messages

	LName	ProviderType
1	Wilson	Physician
2	Carter	Registered Nurse
3	Reed	Physician
4	Nguyen	Physical Therapist
5	Patel	Physician
6	Hernandez	Registered Nurse
7	Kim	Phlebotomist
8	Morris	Physical Therapist
9	Lopez	Physician
10	Brooks	Registered Nurse
11	Sanders	Physician
12	Price	Physical Therapist
13	Bell	Phlebotomist
14	Rivera	Registered Nurse
15	Howard	Physician
16	Ward	Physical Therapist
17	Cox	Physician
18	Peterson	Registered Nurse
19	Flores	Phlebotomist
20	Russell	Physical Therapist
21	Long	Physician
22	Ramirez	Registered Nurse
23	Powell	Phlebotomist
24	Bennett	Physical Therapist
25	Coleman	Physician

11. Create a query that demonstrates a left outer join (includes join in statement)

```sql
Select LName, ProviderType
from Provider P
    left join ProviderType PT --Uses left join
    on P.ProviderTypeID = PT. ProviderTypeID
```

118 % No issues found

Results Messages

	LName	ProviderType
1	Wilson	Physician
2	Carter	Registered Nurse
3	Reed	Physician
4	Nguyen	Physical Therapist
5	Patel	Physician
6	Hernandez	Registered Nurse
7	Kim	Phlebotomist
8	Morris	Physical Therapist
9	Lopez	Physician
10	Brooks	Registered Nurse
11	Sanders	Physician
12	Price	Physical Therapist
13	Bell	Phlebotomist
14	Rivera	Registered Nurse
15	Howard	Physician
16	Ward	Physical Therapist
17	Cox	Physician
18	Peterson	Registered Nurse
19	Flores	Phlebotomist
20	Russell	Physical Therapist
21	Long	Physician
22	Ramirez	Registered Nurse
23	Powell	Phlebotomist
24	Bennett	Physical Therapist
25	Coleman	Physician
26	Smith	NULL
27	Stevens	NULL

12. Create a query that demonstrates a right outer join (includes join in statement)

```sql
Select LName, ProviderType
from Provider P
    right join ProviderType PT --Uses right join
    on P.ProviderTypeID = PT. ProviderTypeID
```

118 % No issues found

Results Messages

	LName	ProviderType
1	Wilson	Physician
2	Reed	Physician
3	Patel	Physician
4	Lopez	Physician
5	Sanders	Physician
6	Howard	Physician
7	Cox	Physician
8	Long	Physician
9	Coleman	Physician
10	Carter	Registered Nurse
11	Hernandez	Registered Nurse
12	Brooks	Registered Nurse
13	Rivera	Registered Nurse
14	Peterson	Registered Nurse
15	Ramirez	Registered Nurse
16	Nguyen	Physical Therapist
17	Morris	Physical Therapist
18	Price	Physical Therapist
19	Ward	Physical Therapist
20	Russell	Physical Therapist
21	Bennett	Physical Therapist
22	Kim	Phlebotomist
23	Bell	Phlebotomist
24	Flores	Phlebotomist
25	Powell	Phlebotomist
26	NULL	CardoVascular Surgeon